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PALEOMAGNETIC EVIDENCE OF A DRIFT OF THE JAPANESE MAIN ISLAND DURING THE PALEOGENE PERIOD

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So far no paleomagnetic study of Paleogene rocks in Japan has been reported. In this paper paleomagnetic results are presented for Paleogene volcanic rocks which were taken from various localities in Southwest Japan. Thermal and alternating field demagnetization experiments have been carefully undertaken to clean the secondary unstable components in the natural remanent magnetization. From the result obtained it is shown that the Paleogene paleomagnetic pole deduced from Southwest Japan is appreciably far from that from the Eurasian continent. This fact might be well accepted as an additional proof supporting a hypothetical drift of the Japanese main island relative to the continent, since a certain time before the Cretaceous period, which was previously reported by the two authors. Such a drift of the Island should still continue during the Paleogene period until the Miocene epoch when the present geographic location of the Island was almost established.

1. INTRODUCTION

Although a great deal of paleomagnetic studies have been made on the Neogene volcanic rocks of Japan, very little work has been performed on the pre-Miocene rocks, especially no Paleogene paleomagnetic result has been reported. Recently, the authors have undertaken a paleomagnetic investigation of Paleogene rocks. A preliminary survey by the authors failed to find reliable fossil remnant magnetization in many Paleogene sedimentary rocks (fairly dominant in Southwest Japan), or in volcanic rock samples of the same age collected from as many localities as possible in Southwest Japan. In Japan, the Paleogene period was geologically one of the most quiet times in volcanic activity in contrast with severe activity in the Neogene period, which might be one of the reasons why as yet no Paleogene paleomagnetic result has been reported from the Japanese region.

2. GEOLOGICAL SETTING

The paleogene volcano-stratigraphy in Southwest Japan has recently been re-examined even though very few data of radiometric dating have been so far available. Various kinds of igneous rocks for paleomagnetism have been collected from Paleogene systems poorly distributed in Southwest Japan. The sampling localities and geologic characters of the samples are shown in fig. 1 and table 1 respectively.

Among the samples concerned in the paper, the age of the Akasaki formation is considered to be lower Eocene (upper *Ypresian-(Cuisian)*), by Asano [1] through his faunal study of globigerinids. On the other hand, rhyolitic welded tuff of upper part of the Futomiyama Group and Okami granodiorite (No. 11, No. 4 in table 1) has been dated at 59 m.y. and 36 m.y. respectively by the K-Ar method [2,3]. The ages of the other rock samples are not so well defined

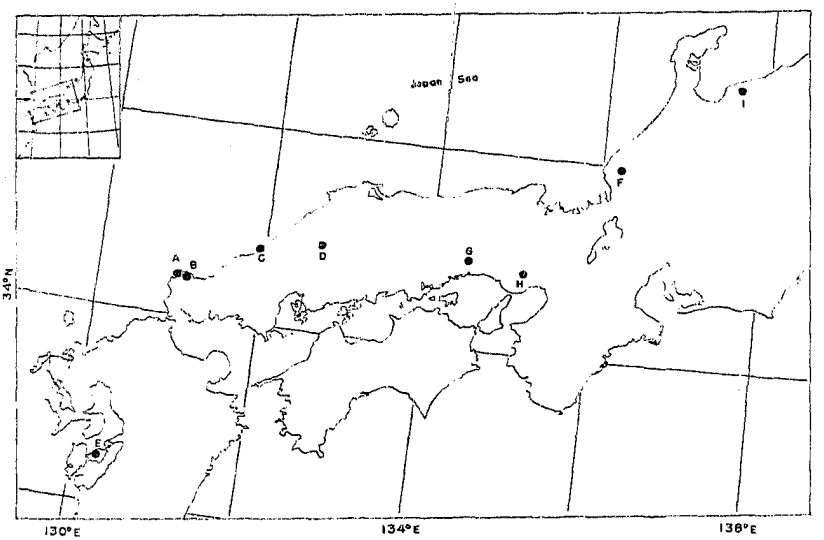


Fig. 1. The index map showing sampling localities in Southwest Japan. A: Yuya-cho, B: Hioki-cho, C: Misumi-cho, D: Kawamoto-cho, E: Kamishima, Amakusa Islands, F: Sabae city, G: Aioi city, H: Kobe city, I: Asahi-cho.

Table 1
List of sampling localities and geologic characters of samples.

No.	Location	Group	Rock unit	Sample		Age
				Site	No.	
1	Hioki-cho Yamaguchi Pref.	Paleogene Volcanics	Augite Andesite	2	8	Upper Paleogene [4]
2	Yuya-cho Yamaguchi Pref.	ditto	Augite Andesite	2	10	ditto
3	Hioki-cho Yamaguchi Pref.	ditto	Hypersthene Olivine Basalt	1	10	ditto
4	Misumi-cho Shimane Pref.	Tamagawa Group	Granodiorite	3	29	36 ± 1.1 m.y. [2]
5	ditto	ditto	Andesite	1	5	Upper Paleogene
6	Kawamoto-cho Shimane Pref.	Kawauchi Volcanic Complex	Quartz Andesite	1	5	Oligocene [5]
7	Amakusa Island Kumamoto Pref.	Akasaki Formation	Red shale	2	11	Upper *Ypresian* (*Cuisian*) [1]

Table 1 (continued)

No.	Location	Group	Rock unit	Sample Site	Sample No.	Age
8	Sabae City Fukui Pref.	Nishitani Group	Rhyolite	2	10	Lower Paleogene [5]
9	Aioi City Hyogo Pref.	Tenkadaiyama Group	Rhyolite	9	46	Lower Paleogene [6]
10	Kobe City Hyogo Pref.	Kongōdōji Group	Rhyolite	2	11	Lower Paleogene [5]
11	Asahi-cho Toyama Pref.	Futomiyama Group	Rhyolitic Welded Tuff	3	13	59 ± 1.8 m.y. [3]

as compared with those mentioned above, and therefore in table 1 they are arranged in order referring to the age estimated from volcano-stratigraphic studies which were published by several geologists [4–6].

3. RESULT OF NRM AND ITS RELIABILITY TEST

Stability examinations of NRM, a.c. – and thermal demagnetization have been made. Considering the directional change before and after a.c. – demagnetization of some test specimens under peak fields up to 400 Oe, most specimens are commonly subjected to alternating peak fields of 100 to 250 Oe. Firstly, an example of a.c. – demagnetization of Tamagawa andesite is illustrated in fig. 2.

In the figure, it is found that fairly good precision has been attained after the magnetic cleaning of peak field 210 Oe. In fig. 3, typical a.c. – demagnetization spectra of Okami granodiorite, Tamagawa andesite and Kawauchi andesite specimens are shown in normalized intensity scale. As a result of magnetic cleaning, all specimens except K-8 showed very little change in remanent direction suggesting a high reliability, whereas specimen K-8 (site 2) showed a large change in remanent direction; for this reason such data are excluded from the present paleomagnetic

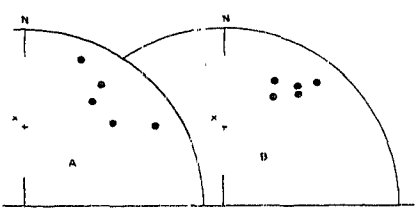


Fig. 2. Directions of NRM before and after the magnetic cleaning in 210 Oe peak field plotted on equal area projections; Tamagawa andesites. A: Initial NRM direction; $k = 25$, B: after magnetic cleaning; $k = 58$.

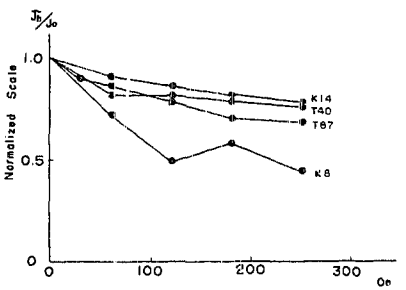


Fig. 3. AC-demagnetization spectra of specimens selected from several rock units. K-8: Kawauchi andesite (site 2), K-14: Kawauchi andesite (site 1), T-40: Tamagawa andesite, T-87: Okami granodiorite.

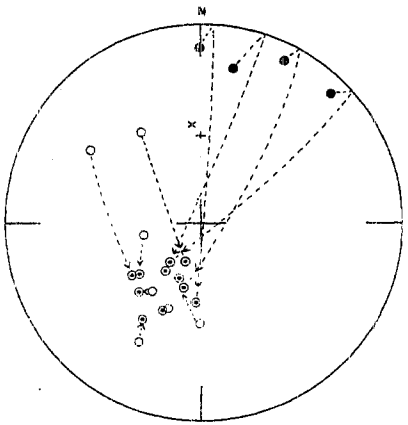


Fig. 4. A prominent increase of grouping in remanent directions after the thermal cleaning at 250°C; Tenkadai-yama rhyolites. ○: upper hemisphere, ●: lower hemisphere, ⊗: after thermal cleaning, upper hemisphere.

results. Secondly, as an example of thermal demagnetization, it is shown that several specimens of Tenkadaiyama rhyolite having scattered NRM grouped together in a consistent reversed direction after heat treatment of 250°C (fig. 4).

The constituent ferromagnetic minerals in the rocks are titanomagnetites with a Curie point of about 500°C and all of our experiments have failed to elucidate self-reversal properties of the rocks. Consequently, this may be the first finding in Japan of evidence suggesting the field reversal during the early Paleogene period.

The results of NRM after partial demagnetizations are given in table 2, together with Fisher's precision parameter k and corresponding virtual pole positions.

The mean value of site means and its corresponding virtual pole position are also given in table 2. The virtual pole positions inferred from each site and their mean are illustrated in fig. 5, along with the Paleogene paleomagnetic pole available from the Eurasian continent.

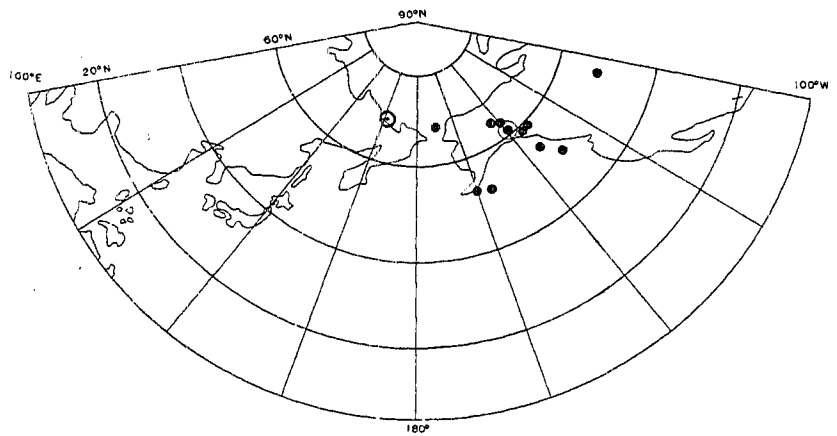


Fig. 5. A map showing comparison between the two different Paleogene paleomagnetic poles computed from the data of Southwest Japan and Eurasia. ●: virtual north pole positions for various sites in Southwest Japan. ○: for Eurasian continent calculated from Irving [7], Kalashnikov [8], Khramov et al. [9], and Lee et al. [10].

Table 2
Results of NRM after magnetic cleaning and virtual pole positions.

No.	Rock unit	No. of samples	NRM		k	α_{95}	VPP	
			Decl.	Incl.			φ	λ
1	Sakamoto Andesite	8	31	56	19	13.1	N66	W146
2	Tsuō Andesite	10	45	63	119	4.5	N54	W160
3	Imamisaki Basalt	10	32	59	36	8.2	N64	W140
4	Okami Granodiorite	29	45	48	25	5.0	N59	W133
5	Tamagawa Andesite	5	33	24	58	10.1	N52	W106
6	Kawauchi Quartz Andesite	5	339	55	60	10.0	N73	E52
7	Amakusa Red Shale	11	43	43	46	7.1	N50	W133
8	Hashidate Rhyolite	10	33	50	72	6.3	N59	W136
9 *	Tenkadaiyama Rhyolite	46	193	−61	34	3.4	N68	W170
10	Kongōdōji Rhyolite	11	40	50	21	10.4	N56	W135
11	Futomiyama Rhyolitic Welded Tuff	13	46	59	18	10.3	N53	W156
Mean **		158	33	53	40	7.1	N62	W139

* Corresponding VPP is calculated irrespective of the sign of NRM direction.
** Data of sample no. 6 are excluded from the mean, because of its exceptional direction of NRM and observed petrologic evidences suggesting intensive hydrothermal alterations.

4. CONSIDERATION

In recent years the authors have undertaken a detailed paleomagnetic investigation of Neogene volcanic rocks in Southwest Japan [11]. According to the results, it can be safely concluded that the dipole axis during this period has maintained a direction nearly parallel to the present Earth's rotation axis in accordance with the results so far obtained by Nomura [12] and Kawai et al. [13].

On the contrary, it may be noticed that the Paleogene paleomagnetic pole for Southwest Japan (139°W, 62°N) is significantly further from either the present geographic pole or the Eurasian paleomagnetic pole (161°E, 69°N) as shown in fig. 5; strictly speaking, of course, the former lies without an oval of 95% confidence level of the latter.

One of the simplest interpretations of the fact mentioned above may be that the discrepancy between the two paleomagnetic poles can be reconciled with a presumption of land drift; that is, a hypothetical dislocation of the Japanese main island relative to the continent. The fact lends further strong support to the reality of the hypothesis previously reported by the two authors [14].

It is needless to say that the geographic reconstruction of an ancient land location cannot be uniquely predicted only from paleomagnetic data. However, a

simplified answer of a possible displacement can be given by the technique of spherical triangulation provided that the central point of land mass traces a route along either the 135°E longitude (orthodrome) or a loxodrome starting from any given longitude, for instance, 125°E. The result is illustrated in fig. 6, in which are schematically drawn the computed isocline and the given course of drift, together with an accompanied rotation of the island during the Paleogene period.

Several years ago, Kawai, Ito and Kume [13] suggested the bending hypothesis of the Japanese island during pre-Tertiary ages as inferred from their result of NRM, which was obtained from two structurally defined sections of the island, i.e. Northeast- and Southwest Japan. They also pointed out a supposed difficulty in the study of pre-Tertiary paleomagnetism concerning the Japanese area.

The present authors, however, do not take such a viewpoint of deformation supposed in the present situation of the Island, but an alternative differential deformation due to an accompanied rotation between the two sections of the Island throughout the drift mentioned above.

In this respect, the present authors' idea does not necessarily deny the bending hypothesis. One of the most plausible accommodations of the two hypotheses may be that throughout the drift of the Japanese main island in the Paleogene period, there accompanied some twenty degrees rotation of Southwest Japan, which would result in the bending of about 40° between Northeast- and Southwest Japan.

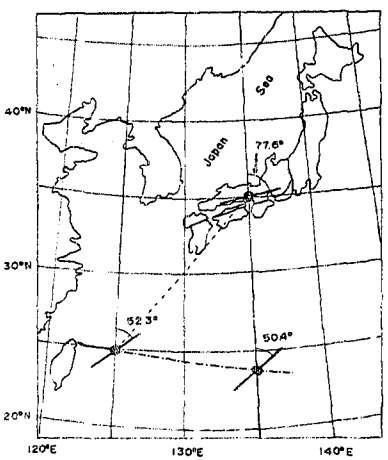


Fig. 6. A schematic view of supposed ancient situations of Southwest Japan, suggesting tentative courses and accompanied clockwise rotations during hypothetical drift of the island.

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